VERA Common Input

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Chapter 1

Introduction

1.1 Introduction to CASL

The Consortium for Advanced Simulation of Light Water Reactors (CASL) is the first DOE Energy Innovation Hub established in July 2010, for the purpose of providing advanced modeling and simulation (M&S) solutions for commercial nuclear reactors.

CASLs vision is to predict, with confidence, the performance of nuclear reactors through comprehensive, science-based modeling and simulation technology that is deployed and applied broadly throughout the nuclear energy industry to enhance safety, reliability, and economics.

CASLs mission is to provide coupled, higher-fidelity, usable modeling and simulation capabilities needed to address light water reactor operational and safety performance-defining phenomena.

CASLs foundational technology products include CASL solutions and CASL ModSim Technologies. CASLs ModSim technology, the Virtual Environment for Reactor Applications (VERA), provides higher-fidelity results than the current industry approach by incorporating coupled physics and science-based models, state-of-the-art numerical methods, modern computational science, integrated uncertainty quantification (UQ) and validation against data from operating pressurized water reactors (PWRs), single-effect experiments, and integral tests.

CASL will address, through new insights afforded by its ModSim technology, key nuclear energy industry challenges to furthering power uprates, higher fuel burnup, and lifetime extension while providing higher confidence in enhanced nuclear safety and this cleaner energy source.

The CASL Team is a consortium that consists of ten core partners and numerous contributing members. The CASL organization is led by Oak Ridge National Lab, and CASLs R&D is

executed in six technical teams called Focus Areas (FA) and one integrating technical area, Collaboration and Ideation (C&I). This ground-breaking partnership provides unparalleled collective institutional knowledge, nuclear science and engineering talent, computational science leadership, and a record of LWR design and regulatory accomplishments!

1.2 VERA Core Simulator

One piece of VERA is the VERA Core Simulator. The core simulator is the specific collection of multi-physics computer codes used to model and deplete a LWR core over multiple cycles. Examples of the separate physics codes include cross section models, neutron transport, depletion, thermal-hydraulics, and fuel performance.

The purpose of the core simulator is to provide data and boundary conditions to model CASL Challenge Problems such as CIPS, CILC, DNB, and PCI analysis.

One important feature of the core simulator is that a single common input file is used to drive all of the different physics codes[1]. One benefit of using a single common input is that users only need to understand and be proficient with one input, instead of having to understand multiple inputs for multiple physics codes. Another benefit of using a single common input is that all codes will be working from a single geometry description and this will reduce errors due to inconsistent models.

This report describes the VERA Common Input file as of October 2013. Additional releases will be made as new features are added in the future.

[1] The only exception to this is for CFD codes, which require a CAD drawing to perform any meaningful analysis.

Chapter 2

User Manual

The VERA input is designed to be modular. The input is split into separate modules (or blocks) to describe different geometric objects in the core, and also separate modules to define specific modeling options for each of the physics codes.

Geometric objects are defined as the physical "parts" of the reactor, which includes fuel assemblies, control rod assemblies, removable burnable poison assemblies, and detectors. By defining each geometric object as a separate module, the objects can be described independently of each other and rely on very little global information. The independent descriptions make QA easier and allows objects to be defined on one cycle and re-used in multiple cycles without worrying about conflicts. Another advantage of the module approach is that it makes it easier to shuffle fuel assemblies, and insert and withdraw "inserts" (such as control rods, detectors, and removable burnable poison assemblies) into the fuel assemblies as the core configuration changes.

Additional modules/blocks are used to define modeling options in each of the physics codes. Separating the geometry description from the modeling options allows all of the physics codes to share the same geometry description and also allows for the same input to be used with multiple physics codes.

A listing of some of the input blocks in the VERA input is:

CASEID This block contains an input title card.

CORE This block describes the core layout including core map, assembly locations, control rod locations, and assembly insert locations. The CORE block contains data that will not change during a cycle depletion.

STATE This blocks describes the statepoint values at a particular point in time. This include inlet temperature, pressure, power, and control rod positions. STATE values can (and

usually do) change at each statepoint.

ASSEMBLY This block contains the physical description of a fuel assembly, not including control rods, detectors, or inserts.

INSERT This block contains the physical description of an assembly insert. An insert is a generic term and can be used to describe a removable burnable poison assembly or a thimble plug assembly.

CONTROL This block contains the physical description of a control rod assembly. A control rod assembly is very similar to an assembly insert, except that it can move during operations.

DETECTOR This block contains the physical description of a detector string.

EDITS This block contains information on what edits the code should produce.

COUPLING This block contains parameters for coupling different physics codes together.

In addition to these blocks listed above, there are additional blocks that contain code-specific information. Examples of these code blocks are **COBRATF**, **INSILICO**, **MAMBA2D**, and **MPACT**. Additional code input blocks can be added as new physics codes are added to the simulator.

The following sections in this chapter describe the most common concepts and features of each input block. A complete listing of all the input cards in each input block is included in Chapter 4.

2.1 Core Description

The [CORE] block describes the core configuration. This block describes the core layout, including where the assemblies are placed, where the control rod locations are, where the removable burnable poison assemblies are inserted, and other core parameters that do not change during a cycle depletion.

The geometric objects inside the core are defined in separate input blocks, the [CORE] block simply describes how all of these objects are placed together.

2.1.1 Core Geometry

The first thing that the code needs to know is the core geometry. The overall *size* of the core is given by the number of assemblies along one major axis of the core. The assembly pitch (*apitch*) defines the width of each assembly, including the assembly gap. The distance from the lower core plate to the upper core plate is given by the *height*. The assembly layout is given by the

core_shape map. Note that the core shape map is the only "square" core map and it must be of *size* assemblies by *size*. Once the core shape is defined, subsequent core maps only have to enter data for actual fuel assembly locations.

```
size 15               ! number of assemblies across one axis
apitch 21.5           ! assembly pitch (cm)
height 406.337        ! distance from lower core plate to upper core plate (cm)

core_shape
  0 0 0 0 1 1 1 1 1 1 1 0 0 0 0
  0 0 1 1 1 1 1 1 1 1 1 1 1 0 0
  0 1 1 1 1 1 1 1 1 1 1 1 1 1 0
  0 1 1 1 1 1 1 1 1 1 1 1 1 1 0
  1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
  1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
  1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
  1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
  1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
  1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
  1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
  0 1 1 1 1 1 1 1 1 1 1 1 1 1 0
  0 1 1 1 1 1 1 1 1 1 1 1 1 1 0
  0 0 1 1 1 1 1 1 1 1 1 1 1 0 0
  0 0 0 0 1 1 1 1 1 1 1 0 0 0 0
```

The *core_shape* map is unique because it is square in shape and composed of the integers 1 and 0. The purpose of this map is to define the shape for the subsequent core maps.

Most physics codes support both calculations run in either full-core symmetry or quarter-core symmetry. If a calculation is run in qtr-core symmetry, the code must know if the symmetry is mirror symmetric or rotational symmetric. This qtr-core symmetry is defined with the *bc_sym* input card. The symmetry option is not used if the calculation is run in full-core symmetry.

```
bc_sym mir    ! define qtr-core symmetry as mirror
```

(Note: rotational symmetry is not supported at this time)

2.1.2 Core Maps

Core maps are used to define the location of geometry objects in the core. The maps are composed of arbitrary length character strings. Even though the character strings can be any size, you usually want to use fairly compact names so the maps remain legible.

All of the maps require one entry for each assembly location defined in the *core_shape* map. However, the input parser can be used to take advantage of core symmetry. If the core is symmetric, you only need to input the maps in quarter or octant symmetry, and the input parser will automatically fold the map out to full-symmetry. (Enter the SE quadrant for qtr-symmetry and the SEE octant for octant-symmetry.) The method used to input the core maps is independent of how the actual calculations are run. For example, you can enter all the core maps in octant symmetry and still run the calculations in quarter or full symmetry.

If there is an empty location in the map (e.g. if there is no detector or no control rod in an assembly), enter a dash "-". The dash is significant and signifies a blank location in the core map.

The *assm_map* shows where the assembly types are located within the core. In the example below, there are three assembly types which will be defined in [ASSEMBLY] block(s).

```
assm_map
                A3 A3 A3 A3 A3 A3 A3
           A3 A3 A3 A1 A3 A1 A3 A1 A3 A3 A3
        A3 A3 A2 A1 A2 A1 A2 A1 A2 A1 A2 A3 A3
        A3 A2 A2 A2 A1 A2 A1 A2 A1 A2 A2 A2 A3
     A3 A3 A1 A2 A1 A2 A1 A2 A1 A2 A1 A2 A1 A3 A3
     A3 A1 A2 A1 A2 A1 A2 A1 A2 A1 A2 A1 A2 A1 A3
     A3 A3 A1 A2 A1 A2 A1 A2 A1 A2 A1 A2 A1 A3 A3
     A3 A1 A2 A1 A2 A1 A2 A1 A2 A1 A2 A1 A2 A1 A3
     A3 A3 A1 A2 A1 A2 A1 A2 A1 A2 A1 A2 A1 A3 A3
     A3 A1 A2 A1 A2 A1 A2 A1 A2 A1 A2 A1 A2 A1 A3
     A3 A3 A1 A2 A1 A2 A1 A2 A1 A2 A1 A2 A1 A3 A3
        A3 A2 A2 A2 A1 A2 A1 A2 A1 A2 A2 A2 A3
        A3 A3 A2 A1 A2 A1 A2 A1 A2 A1 A2 A3 A3
           A3 A3 A3 A1 A3 A1 A3 A1 A3 A3 A3
                 A3 A3 A3 A3 A3 A3 A3
```

The following map is equivalent to the previous map, but demonstrates a map with octant symmetry. Only values in the SEE octant are entered and the parser will automatically fold the map out to full-symmetry.

```
assm_map
   A1
   A2 A1
   A1 A2 A1
   A2 A1 A2 A1
   A1 A2 A1 A2 A2
   A2 A1 A2 A1 A2 A3
   A1 A3 A1 A3 A3 A3
```

```
    A3 A3 A3 A3          ! assembly map with octant symmetry
```

The *insert_map* is used to show where assembly inserts are located within the core. In the following example, the inserts are burnable poison assemblies with different numbers of pyrex rods. The *insert_map* can also be used to define thimble plugs locations. The geometry description of the inserts will be given in the [INSERT] block. The *insert_map* is optional if no inserts are present in the core.

```
insert_map
    -    BP20   -    BP20   -    BP20   -    BP12
   BP20   -    BP24   -    BP20   -    BP24   -
    -    BP24   -    BP20   -    BP16   -    BP8
   BP20   -    BP20   -    BP20   -    BP16   -
    -    BP20   -    BP20   -    BP24   -
   BP20   -    BP16   -    BP24  BP12   -
    -    BP24   -    BP16   -     -
   BP12   -    BP8    -
```

The *det_map* is used to shows where the detectors are located. The geometry description of the detector strings is given in the [DETECTOR] block. In this example, there is only one detector type denoted with a "1". Since the "1" occurs in a core map, it is treated as a character string. The *det_map* is optional if no detectors are present in the core.

```
det_map
                - - 1 - - 1 -
          1 - - 1 - 1 - - - - - -
        - - - - - - 1 - 1 - 1 - 1
        1 1 - - - - 1 - - - - - -
    - - - - 1 - - - 1 - 1 - 1 - -
    1 - 1 - - 1 - 1 - - - - - 1 -
    - - - 1 - - 1 - - 1 - - 1 - -
    1 - 1 - 1 - 1 - - 1 - 1 1 1 -
    - 1 - - - - - - 1 - 1 - - - 1
    - - - - 1 - 1 - - - - 1 - - -
    1 - - - 1 - - 1 - - 1 - - - 1
       - - - - 1 - - 1 - - 1 - -
       - 1 - 1 - - 1 - - - - - 1
          1 - - - 1 - - 1 - 1 -
                1 - - 1 - - -
```

The control rod assemblies are described with two maps. The *crd_map* defines the control rod types and locations in the core. The *crd_bank* map assigns control rod locations to control rod banks. The control rod maps are optional if no control rods are present in the core.

```
crd_map
                - - - - - - - -
          - 1 - 1 - 1 - 1 - 1 - 1 -
      - - - 1 - 1 - 1 - 1 - 1 - - -
      1 - 1 - - - 1 - - - 1 - 1
   - - 1 - 1 - - - - - - 1 - 1 - -
   - 1 - - - 1 - 1 - 1 - - - 1 -
   - - 1 - - - - - - - - - 1 - -
   - 1 - 1 - 1 - 1 - 1 - 1 - 1 -
   - - 1 - - - - - - - - - 1 - -
   - 1 - - - 1 - 1 - 1 - - - 1 -
   - - 1 - 1 - - - - - 1 - 1 - -
      1 - 1 - - - 1 - - - 1 - 1
      - - - 1 - 1 - 1 - 1 - - -
         - 1 - 1 - 1 - 1 - 1 -
               - - - - - - -
crd_bank
               - - - - - - -
         - SA - B - C - B - SA -
      - - - SD - SB - SB - SC - - -
     SA - D - - - D - - - D - SA
   - - SC - A - - - - - A - SD - -
   - B - - - C - A - C - - - B -
   - - SB - - - - - - - - - SB - -
   - C - D - A - D - A - D - C -
   - - SB - - - - - - - - - SB - -
   - B - - - C - A - C - - - B -
   - - SD - A - - - - - A - SC - -
     SA - D - - - D - - - D - SA
      - - - SC - SB - SB - SD - - -
         - SA - B - C - B - SA -
               - - - - - - -
```

2.1.3 Core Plates and Baffles

The upper and lower core plates are defined with a material composition, a thickness, and a volume fraction of the structural material. The remainder of the volume fraction is filled with coolant.

The core baffle (or shroud) is defined with a material, the size of the gap between the outer assembly and baffle, and the baffle thickness.

```
lower_plate SS304  5.0  0.5   ! material, thickness (cm), volume fraction
upper_plate SS304  7.6  0.5   ! material, thickness (cm), volume fraction
baffle      SS304 0.19 1.26   ! material, gap (cm), and thickness (cm)
```

The core plates are modeled in the neutronics codes as smeared materials. This is a reasonable approximation since the plates are relatively far away from the active fuel and are mostly composed of steel and water. The user specifies a plate height and a volume fraction of the plate material. (physically, the core plate is usually a large steel plate with flow holes passing through it.) The remainder of the volume fraction in the core plate is filled with coolant. The coolant density is updated during the iterations with the local T/H conditions.

2.1.4 Small Cores Geometries

Even though the VERA input is designed for "real" core geometries, it can accommodate smaller problems as well. For example, if you only want to run a single-assembly calculation, you would define the core size as one assembly by one assembly, and all of the core maps would contain a single assembly.

```
size 1                ! core composed of a single-assembly
core_shape
   1
```

If you want to model a single fuel rod, you would define a core with one assembly in it and an assembly with one rod in it.

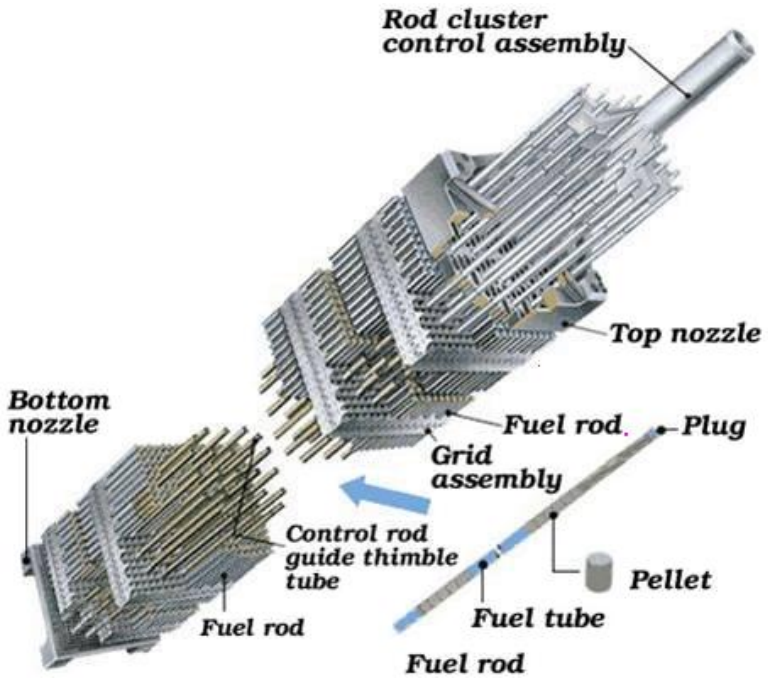


Figure 2.1: PWR Fuel Assembly (source: Public internet).

2.2 Assembly Description

The [ASSEMBLY] block contains the geometric description of a unique fuel assembly. Multiple [ASSEMBLY] blocks are allowed to described different assembly designs in the core.

If there are multiple assembly designs that are geometrically similar (i.e. everything is the same except enrichments), then they can all be defined in a single [ASSEMBLY] block. Each assembly type will have a unique *axial* card with possibly unique axial levels and lattice types. Assemblies within a single reload typically have a design similar enough that they can share a single [ASSEMBLY] block.

If assembly designs are sufficiently different (e.g. different vendors, different generations, etc.) they need to be defined with their own [ASSEMBLY] block. Having different blocks for each assembly design allows the blocks to be defined and archived independently.

A typical PWR assembly is shown in Figure 2.1. Refer to this figure in the following discussions.

2.2.1 Initial Data

Each assembly must contain a geometry description with the number of pins across the assembly and the pin pitch.

```
title "Westinghouse 17x17"      ! assembly title
npin 17                         ! number of pins across one side
ppitch 1.260                    ! pin pitch (cm)
```

The number of pins *npin* must be the same for every assembly in a core.

The inter-assembly gap on each side of the assembly is calculated with $[apitch - npin * ppitch]/2$

The fuel and structural materials are defined with the following cards. See Chapter 3 for a complete description of the material inputs.

```
fuel U31 10.257 95.0 / 3.1   ! mat, density (g/cc), Theoretical density (%)
                             !    / U-235 enrichment (%)
mat inc  8.19                ! mat, density (g/cc)
mat ss   8.0
mat zirc 6.56
```

2.2.2 Cell Descriptions

Cell cards are used to describe a pincell. The first parameter on the *cell* card is the cell ID. This is followed by a list of radii for each ring in the pincell and a slash. After the slash is the list of materials that go into each ring. The cell ID's are used in the lattice descriptions described in the next section.

```
cell 1     0.4096 0.418 0.475 / U31 he zirc
cell GT           0.561 0.602 / mod    zirc       ! guide tube
cell IT           0.561 0.602 / mod    zirc       ! instrument tube
cell 7            0.418 0.475 / mod    mod        ! empty location
cell 8            0.418 0.475 /     he zirc       ! plenum
cell 9                  0.475 /        zirc       ! pincap
```

In this example, in cell 1, the material "U31" extends from radius 0 to 0.4096. The material "he" extends from a radius 0.4096 to 0.418. The materials "U31" and "he" are defined on *fuel* and *mat* cards, respectively.

The outside of each pin is automatically filled with the special material "mod". The composition of "mod" is determined by the local T/H conditions and the boron concentration. The moderator cannot be specified on a *mat* card.

2.2.3 Lattice Descriptions

Once the cells are defined, they are are placed into 2D "lattices" as shown below. Like the core maps, the lattice maps can be entered with either full-symmetry, qtr-symmetry, or octant-symmetry. The maps below are octant symmetric maps for 17x17 assembly designs.

```
lattice FUEL1
    IT
     1 1
     1 1 1
    GT 1 1 GT
     1 1 1  1 1
     1 1 1  1 1 GT
    GT 1 1 GT 1  1 1
     1 1 1  1 1  1 1 1
     1 1 1  1 1  1 1 1 1

lattice LGAP1
    IT
     7 7
     7 7 7
    GT 7 7 GT
     7 7 7  7 7
     7 7 7  7 7 GT
    GT 7 7 GT 7  7 7
     7 7 7  7 7  7 7 7
     7 7 7  7 7  7 7 7 7

lattice PLEN1
    IT
     8 8
     8 8 8
    GT 8 8 GT
     8 8 8  8 8
     8 8 8  8 8 GT
    GT 8 8 GT 8  8 8
     8 8 8  8 8  8 8 8
     8 8 8  8 8  8 8 8 8
```

```
lattice PCAP1
    IT
     9 9
     9 9 9
    GT 9 9 GT
     9 9 9  9 9
     9 9 9  9 9 GT
    GT 9 9 GT 9  9 9
     9 9 9  9 9  9 9 9
     9 9 9  9 9  9 9 9 9
```

Lattice cards define each unique axial level in the assembly. The first parameter is the lattice name (e.g. FUEL1, PCAP1, etc.), followed by a map of the *cell* ID's.

2.2.4 Axial Descriptions

After lattice maps are defined for each axial level, the lattices are "stacked" into an assembly using *axial* cards as shown below.

```
axial A1    6.050
    LGAP1  10.281
    PCAP1  11.951
    FUEL1 377.711
    PLEN1 393.711
    PCAP1 395.381
    LGAP1 397.501
```

The *axial* card tells the code how to place the lattices axially. The first parameter is the name of the assembly (A1), followed by a list of elevations and lattice types. For example, lattice "FUEL1" extends from 11.951 to 377.711 cm axially.

If multiple assembly types are defined in one [ASSEMBLY] block, there will be multiple *axial* cards, each with a unique assembly ID.

All elevations are defined relative to the lower core plate.

2.2.5 Grid Spacer Descriptions

Grid cards are used to define unique grid spacer types. The following example defines to grid types, "END" and "MID".

```
grid END inc  1017  3.866   ! grid spacer material, mass(kg), height(cm)
grid MID zirc  875  3.810
```

The grid types are placed axially with the *grid_axial* card:

```
grid_axial
    END  13.884
    MID  75.2
    MID 127.4
    MID 179.6
    MID 231.8
    MID 284.0
    MID 336.2
    END 388.2
```

The elevations are the midpoints of the spacer grid and are relative to the lower core plate.

2.2.6 Nozzle Descriptions

The assembly nozzles are modeled in the neutronics codes as smeared materials. This is a very good approximation since the nozzles are not in the active fuel region and are mostly composed of water, steel, and zirconium. The user only specifies a nozzle mass and a nozzle height. The total volume of the nozzle region is calculated from the assembly pitch and nozzle height. The volume of the nozzle is calculated from the nozzle mass and density. The volume of the coolant is then calculated as the total volume minus the volume of the nozzle. The coolant density is updated with the local T/H conditions.

```
lower_nozzle  ss 6.05  6250.0  ! mat, height, mass (g)
upper_nozzle  ss 8.827 6250.0  ! mat, height, mass (g)
```

Note that the *lower_nozzle* height should match the bottom elevation on the *axial* card. The *upper_nozzle* height + the top elevation on the *axial* card must match the core *height* in the [CORE] block.

2.3 Control Rod Assembly Description

The [CONTROL] block contains the geometric description of a control assembly.

A control rod assembly is defined in the same way as a fuel assembly is described. You define cells, lattices, and axial descriptions. The only difference between the control rod assembly and the fuel assembly is that the control rod assembly describes what is inside the guide tubes and the assembly description defines the guide tubes themselves.

The geometric description for a control rod is given in the **fully inserted** position.

```
title "B4C control rods with AIC tips"   ! control_title
npin 17                                  ! control_num_pins

cell 1  0.382 0.386 0.484 / aic he ss      ! AIC cell
cell 2  0.373 0.386 0.484 / b4c he ss      ! B4C cell

rodmap  AIC
   -
   - -
   - - -
   1 - - 1
   - - - - -
   - - - - - 1
   1 - - 1 - - -
   - - - - - - - -
   - - - - - - - - -

rodmap  B4C
   -
   - -
   - - -
   2 - - 2
   - - - - -
   - - - - - 2
   2 - - 2 - - -
   - - - - - - - -
   - - - - - - - - -

axial CR1 15.76 AIC  376.44 B4C  394.3
```

Note that the name of the control rod "CR1" refers to the control rod type in the *crd_map* in the [CORE] block.

Note that the locations of the control rod fingers must match guide tube locations in the corresponding [ASSEMBLY] block descriptions. Furthermore, the outer radii of the control rod fingers must be smaller than the inner radii of the guide tubes.

The user can define materials in the [CONTROL] block. These materials only have scope in this

block and are not accessible by other blocks. See Chapter 3 for details.

2.3.1 Control Rod Stroke

One difference between the control rod description and the assembly description is that the control rod moves during operation and the movement must be defined with a *stroke* card.

The first value on the *stroke* card is the total length of the control rod travel (stroke) from fully inserted to fully withdrawn.

The second value on the *stroke* card is the number of steps in the fully withdrawn position. Step 0 is the fully inserted position.

```
stroke  360.0 228      ! stroke (cm), number of steps fully withdrawn
```

All elevations given provided in the geometric description are for a control rod in the fully inserted position (step 0).

2.3.2 Position Example

From the *axial* card shown above, the bottom of the AIC at the fully-inserted position is 15.46 cm. From the *stroke* card, the total stroke is 360.0 cm and the number of steps in the fully withdrawn position is 228 steps. Therefore, the bottom elevation of the AIC lattice at step N will be

$$E(N) = 15.46 + \frac{360.0 \cdot N}{228} \tag{2.3.1}$$

Using this formula, the bottom elevation of the AIC lattice at the following step positions is:

- step 228 (fully withdrawn)= 15.46 + 360.0 * 228 / 228 = 375.46 cm
- step 100 = 15.46 + 360.0 * 100 / 228 = 173.35 cm
- step 0 (fully inserted) = 15.46 + 360.0 * 0 / 228 = 15.46 cm

2.4 Insert Description

An assembly insert is defined in the same way as a fuel assembly or control rod assembly is described. You define the insert using cells, lattices, and axial descriptions.

The only difference between an assembly insert and the fuel assembly description is that assembly insert describes what is inside the guide tubes and the assembly description defines the guide tubes themselves.

The following example shows a definition of a Pyrex insert.

```
[INSERT]
    title "Pyrex"
    npin 17
    mat pyrx1 2.25 pyrex-vera
    cell 1  0.214 0.231 0.241 0.427 0.437 0.484 / he ss he pyrx1 he ss
    rodmap  PY24
       -
       - -
       - - -
       1 - - 1
       - - - - -
       - - - - - 1
       1 - - 1 - - -
       - - - - - - - -
       - - - - - - - - -

    axial INS24  15.76 PY24 376.441
```

The name of the insert "INS24" refers to an insert type defined in the *insert_map* in the [CORE] block.

The locations of the insert fingers must match the guide tube locations in the corresponding [ASSEMBLY] block descriptions. Furthermore, the outer radii of the insert fingers must be smaller than the inner radii of the guide tubes.

2.5 Detector Description

A detector string is defined in the same way that a fuel assembly is described. You define cells, lattices, and axial descriptions to define the detector string.

```
[DETECTOR]
  title "Incore instrument thimble"
  npin 17

  mat he 0.0001786
```

```
mat ss 8.0

cell 1  0.258 0.382 / he ss

rodmap  LAT
   1
   - -
   - - -
   - - - -
   - - - - -
   - - - - - -
   - - - - - - -
   - - - - - - - -
   - - - - - - - - -

axial D1  0.0 LAT 406.337
```

2.6 Statepoint Values

The [STATE] block defines the state of the core at a particular point in time (power, flow, pressure, inlet temperature, rod positions, boron concentration, etc.). These values will typically change during a depletion.

An example showing the most common input cards in the [STATE] block is shown below.

```
[STATE]
  power   98.0       ! % of rated power
  flow   100.0       ! % of rated flow
  pressure 2250.0    ! psia
  tinlet 557.33      ! F
  feedback on        ! turn on T/H feedback

  boron  1285        ! initial boron ppmB
  search 'boron'     ! turn on boron search

  sym qtr            ! run problem in qtr-symmetry

  rodbank SA 228
          SB 228
          SC 228
          SD 228
           A 228
           B 228
           C 228
           D 167
```

The *sym* card tells the code to run the calculation in full-core symmetry or qtr-core symmetry. If the calculation is run in qtr-core symmetry, the symmetry is either set to qtr-core rotational or qtr-core mirror by the *bc_sym* card in the [CORE] block.

The *rodbank* card is used to position the control rods. The input includes pairs of bank names and bank positions. The bank names correspond to the *crd_map* in the [CORE] block. The positions indicate the position of the control rod bank in steps. For most PWR's, 0 steps is fully inserted. The number of steps for a rod to be completely withdrawn is set by the *stroke* card in the [CONTROL] block. For Westinghouse PWR's, a typical value of fully-withdrawn is 228 steps.

2.7 Edits

The [EDITS] block is used to define the edits the user would like to see.

The *axial_edit_bounds* defines the axial levels to average the power over. The user may choose to average power over uniform axial intervals (like most nodal codes), or to specify the edit intervals manually.

2.7.1 CTF Nodalization

The [EDITS] block is also used to set the axial nodalization in CTF.

If running CTF, there is a restriction that the grid boundaries must be explicitly included in the *axial_edit_bounds*. This can get a little complicated for the user. The grids are defined by specifying the grid heights on the *grid* card and the elevations of the grid midpoints on the *grid_axial* card. This heights and midpoints must be converted to the elevations at the top and bottom of the grid, and then the top and bottom elevations must be included in the *axial_edit_bounds*.

For example, if a grid is defined with a centerline at 75.0 and a height of 2.5, then the *axial_edit_bounds* must include the points $75.0 \pm 1.25 = 73.75$ and 76.25.

```
[ASSEMBLY]
 grid GRID1 inc 1000 2.5    ! grid name, material, mass (g), and height (cm)
 grid_axial                 ! locations of grid midpoints (cm)
     GRID1 75.0

[EDITS]
 axial_edit_bounds
    ...
    73.75                   ! this array must include top and bottom grid boundaries
    76.25
    ...
```

The reason for this restriction is because the power is already calculated on the *axial_edit_bounds*, so it is natural to use these boundaries in the CTF model as well. The grids must be explicitly included in the CTF boundaries so the loss coefficients are calculated correctly.

In the future, this restriction may be lifted and an additional edit bounds array will be added explicitly for CTF calculations.

2.8 Coupling Parameters

The [COUPLING] block defines the relaxation parameters and convergence criteria to be used when coupling different physics codes. These values are used to determine convergence *between* physics codes. Convergence criteria *within* a physics code is controlled by the specific code block.

Refer to Chapter 4 for a complete listing of all the cards in the [COUPLING] block.

No code-specific information is included in the [COUPLING] block. All code-specific information is contained in the code-specific blocks.

As an example, consider the following multi-physics code coupling:

1. Run T/H calculation
2. Run neutronics calculation
3. Check eigenvalue convergence
4. Check power convergence
5. Relax/dampen the power shape
6. If not converged, go to step 1.

The eigenvalue convergence in step 3 uses the parameter *epsk* to check the change in eigenvalue between coupled iterations. There are additional eigenvalue convergence criteria *within* the neutronics code, but the internal parameters are specified in the individual code blocks.

The power convergence in step 4 uses the parameter *epsp* to check the change in power between coupled iterations.

Additional convergence checks are made on the peak fuel temperature, maximum change in density, and change in boron concentration (if applicable).

The example shown above uses a Picard iteration to converge. Picard iterations usually need to apply a relaxation factor (also called a damping factor or under-relaxation factor) to one or more of the calculated quantities to converge. The relaxation factors are applied in the following manner:

$$x = \omega\, x^{\text{new}} + (1-\omega) x^{\text{old}} \tag{2.8.1}$$

where x is the calculated parameter and ω is the relaxation factor. A relaxation factor of 1.0 signifies no relaxation is performed. A relaxation factor < 1.0 signifies under-relaxation.

Relaxation factors can be specified for the point-wise power, point-wise temperature, and/or point-wise density.

Chapter 3

Materials Description

This chapter contains a description of the material input. There are two types of materials in the input file – structural materials (input with a *mat* card) and fuel materials (input with a *fuel* card).

Structural materials can be defined in either the [CORE] block, or in the geometry object blocks [ASSEMBLY], [INSERT], [CONTROL], and [DETECTOR]. If the materials are defined in the [CORE] block, they have global scope. If the materials are defined in the geometry object blocks, they only have scope in the block they are defined. The reason for this is to maintain the modularity of the geometry objects.

Fuel materials can only be defined in [ASSEMBLY] blocks.

Materials are used in many different input cards. They are used to define cells, nozzles, core plates, baffles, grids, reflectors, etc. Every material that is used must be defined with either a *mat* card or a *fuel* card.

3.1 Structural Materials

Structural materials are not fuel and do not deplete. Structural materials are defined with the following input card:

mat *user-mat density* (*library-name*$_i$, *frac*$_i$, i=1, I)

where:

- *user-mat* is a user-defined material name. It is case sensitive. *user-mat* is used to define material names in other input cards such as *cell*, *grid*, *nozzle*, etc. (No default)
- *density* is the material density in g/cc (No default)
- *library-name* is a corresponding material library name for the user material. The library name must be defined in the material composition file. (Default = *user-mat*). Multiple library materials can be mixed to form a single user material.
- *frac* is the fraction of the library material in the user material. (Default=1.0 if there is only one library material in the user material).

There are two special user materials, "mod" and "vacuum". The user can use these materials in cell definitions, but the code will be automatically determine the composition of these materials based on T/H feedback and soluble boron concentrations. The user is not allowed to define a user materials named "mod" or "vacuum" on a *mat* card.

Some example material cards are shown below.

```
mat zirc 6.56                ! use default library name and fraction
mat zirc 6.56 zirc 1.0
mat B10  12.0 boron 1.0
mat XYZ  6.0  zirc 0.8 ss 0.2
mat ABCD 8.0  zirc 0.8 ss 0.15 b4c 0.05
```

All of the material fractions must sum to either +1.0 or -1.0. If positive fractions are used, the fractions refer to atomic fractions. If negative fractions are used, the fractions refer to weight fractions.

3.1.1 Search Order

Structural materials can be defined in either the [CORE] block or one of the geometry object blocks. When a material is referred to in a block, it will look for the material definition in the following order:

1. The code will first look for the material name in the local block ([ASSEMBLY], [INSERT], [CONTROL], or [DETECTOR])
2. If the material is not found in the local block, it will look in the [CORE] block

If materials are defined in the [CORE] block, they have global scope over the entire input. If materials are defined in other blocks, they only have scope over the local block. This means

that two geometry object blocks can use different material definitions with the same name. One example of this is that two assemblies can be defined with the material "zirc", but the "zirc" can have slightly different compositions in the blocks.

3.2 Composition File

All library materials must be included in the HDF composition file. The name of the composition file is specified on the [INSILICO] *mat_library* card, or the [MPACT] *(TBD)* card, and is usually named something like "casl_comp.sh5".

A partial list of materials that are present on the current composition file is:

- aic
- b4c
- boron
- inc
- he
- mod
- ss
- zirc

3.3 Fuel Materials

Fuel materials are composed mostly of UO_2 with different U-235 enrichments. Fuel materials are different from structural materials in that they deplete and have additional properties as described below.

Fuel can only be defined in [ASSEMBLY] blocks, and fuel materials can only be referenced by *cell* cards in the [ASSEMBLY] block they are defined in.

Fuel materials are defined with the following input card:

fuel *user-mat density thden / U-235_enrichment* {$HM_material_i$=$HM_enrichment_i$, i=1, N}
{ / *gad_material=gad_fraction* }

Where:

- *user-mat* is a user-defined fuel name. It is case sensitive. (No default)
- *density* is the fuel material density in g/cc (No default)
- *thden* is the percent of theoretical density in the pellet (%) (No default)
- *U-235_enrichment* is the U-235 enrichment in the fuel in weight % (No default).
  - If U-234 and U-236 are not specified, the enrichments will automatically be added to the fuel by a pre-determined function (see below)
  - If the sum of the heavy metal (HM) enrichments does not equal 100%, the remainder of the HM composition will be assigned to U-238.
- *HM_material*$_i$ is the material name for a HM (Pu-239, Pu-241, etc.) (optional)
- *HM_enrichment*$_i$ is the enrichment of HM isotope i in weight % (optional)
- *gad_material* is the material name for the gad oxide (or other material) (optional). The gad material may be a mixture defined on a separate *mat* card.
- *gad_fraction* is the weight percent of the gad material relative to the total mass (optional)

Oxygen should not be included on the *fuel* card. The correct amount of oxygen will automatically be added to the HM to create an oxide (either UO_2 or $(HM)O_2$).

The *density* is the "stack density" or "smeared density" and should include the volume of the pellet dishing and chamfers. It is calculated as the total mass of the fuel pellets divided by the total volume of the fuel

$$\text{stack density} = \frac{\text{(fuel mass)}}{\pi(\text{pellet radius})^2\,\text{(fuel height)}} \tag{3.3.1}$$

The *thden* refers to the actual density of the pellet. This quantity is only used in fuel performance codes to evaluate material properties.

If U-234 or U-236 enrichments are not included, they are automatically added to the fuel with the following formulas:

$$W_{234} = 0.007731 \cdot W_{235}^{1.0837} \tag{3.3.2}$$

$$W_{236} = 0.0046 \cdot W_{235} \tag{3.3.3}$$

Where W_{23x} is the enrichment of each of the uranium isotopes in percent.

Examples of valid fuel cards are:

```
fuel U21 10.4 95.2 / 2.1
fuel UO2-35 10.297 95.0 / 3.5
fuel U23 10.111 / 2.3                    ! bare minimum input for fuel
```

```
mat gad5 7.407 gd2o3 1.0                    ! define gad material
fuel U49 10.111 94.5 / 1.8  / gad5=0.05     ! 5% gad
fuel PX1 10.111 94.5 / 0.711  Pu-239=23.4  Pu-241=1.9     ! Sample MOX
fuel UPX21G 10.111 94.5 / 2.1 U-234=0.001 Pu-239=3.4  Pu-241=1.9 / gad3=0.03
```

Chapter 4

Input Card Descriptions

This chapter contains a complete listing of the available input cards.

The input for each block is given in a separate subsection.

Input cards are given in **bold** text followed by the parameters on the card. Following each input card is a description of the parameters on that card.

An example of an input card is:

card_name param1, param2

param1 The first parameter on card (units)

param2 The second parameter on card (units)

4.1 Block CASEID

[CASEID]

title case_id

case_id	Problem name

4.2 Block STATE

[STATE]

title title

title	Statepoint title

power power

power	Operating power (percent of rated power)

flow flow

flow	Operating flow (percent of rated flow)

bypass bypass

bypass	Bypass flow (percent of rated flow)

tinlet tinlet

tinlet	Inlet temperature (degrees F**)

tfuel tfuel

tfuel	Fixed fuel temperatures (degrees K) - Only used without T/H feedback

modden modden

modden	Fixed moderator density (g/cc) - Only used without T/H feedback

xenon xenopt

xenopt	Xenon option (“zer”, “dep”, or “equil”)

samar sampot

sampot	Samarium option (“zer”, “dep”, “equil”, or “peak”)

boron boron

boron	Soluble boron concentration (ppm)

kcrit kcrit

kcrit Critical eigenvalue used in boron search

search search

search Search option ("keff" or "boron")

pressure pressure

pressure Core exit pressure (psia)

deplete deplete

deplete List of depletion exposure steps (MWD/MTU) - only used with depletion

rodbank bank_pos, bank_labels

bank_pos Steps withdrawn for each bank in list

bank_labels List of control rod banks to position. Labels correspond to crd_map in CORE block.

feedback feedback

feedback Flag to turn on and off T/H feedback ("on" or "off"). Currently not used, feedback is controlled by executable that is run.

sym sym

sym Core fraction to run problem in ("full" or "qtr")

4.3 Block CORE

[**CORE**]

name core_name

core_name Name of the reactor core

cycle cycle_num, cyc_stdate, cyc_edate

cycle_num Cycle number

cyc_stdate Cycle startup date (YYYYMMDD)

cyc_edate Cycle end date (YYYYMMDD). Enter zero if end date is not known.

size core_size

core_size Number of assemblies across one axis in full-core geometry (required)

rated rated_power, rated_flow

rated_power Rated thermal power at 100% power (MW)

rated_flow Rated vessel flow at 100% flow (Mlbs/hr**)

apitch apitch

apitch Assembly pitch (cm)

baffle baffle_mat, baffle_gap, baffle_thick

baffle_mat Baffle material

baffle_gap Gap between outside assembly (including assembly gap) and baffle (cm)

baffle_thick Thickness of baffle (cm)

vessel vessel_mats, vessel_radii

vessel_mats Vessel materials

vessel_radii Vessel radii (cm)

core_shape shape

shape Square map showing the fuel assembly locations. Enter 1 for fuel assembly locations and 0 for empty locations.

assm_map assm_map

assm_map Core map of the fuel assembly types. The assembly types correspond to assembly labels in the ASSEMBLY block. All fuel assemblies must have a type defined.

insert_map insert_map

insert_map Core map of the fuel insert types and locations. The insert types correspond to insert labels in the INSERT block. Use a dash to specify assemblies with no inserts.

det_map det_map

det_map Core map of the detector types and locations. The detector types correspond to detector labels in the DETECTOR block. Use a dash to specify assemblies with no detectors.

crd_map crd_map

crd_map Core map of the control rod types and locations. The control rod types correspond to control rod labels in the CONTROL block. Use a dash to specify assemblies with no control rods.

crd_bank crd_bank

crd_bank Core map of the control rod bank labels. These labels are used to position groups of control rods by bank label. Use a dash to specify assemblies with no control rods.

lower_plate lower_mat, lower_thick, lower_vfrac

lower_mat Lower core plate material

lower_thick Lower core plate thickness (cm)

lower_vfrac Lower core plate volume fraction. Remainder of volume fraction will be filled with coolant.

upper_plate upper_mat, upper_thick, upper_vfrac

upper_mat Upper core plate material

upper_thick Upper core plate thickness (cm)

upper_vfrac Upper core plate volume fraction. Remainder of volume fraction will be filled with coolant.

bc_sym bc_sym

bc_sym Symmetry flag for the core when using qtr-symmetry. Flag is not used in full-symmetry. Valid options are "rot" and "mir".

bc_bot bc_bot

bc_bot Bottom neutron transport boundary condition, "vacuum" (default) or "reflecting".

bc_top bc_top

bc_top Top neutron transport boundary condition, "vacuum" (default) or "reflecting".

bc_rad bc_rad

bc_rad Radial neutron transport boundary condition, "vacuum" (default) or "reflecting".

xlabel xlabel

xlabel List of 2-character assembly position labels in x-direction. These values are used in the edit maps.

ylabel ylabel

ylabel List of 2-character assembly position labels in y-direction. These values are used in the edit maps.

height height

height Total axial distance from bottom core plate to upper core plate (cm). Distance does not include core plate thicknesses.

mat mat

mat See detailed Materials description.

lower_ref lower_refl_mats, lower_refl_thicks, lower_refl_vfracs

lower_refl_mats Lower reflector materials

lower_refl_thicks Lower reflector thicknesses (cm)

lower_refl_vfracs Lower reflector volume fractions

upper_ref upper_refl_mats, upper_refl_thicks, upper_refl_vfracs

upper_refl_mats Upper reflector materials

upper_refl_thicks Upper reflector thicknesses (cm)

upper_refl_vfracs Upper reflector volume fractions

4.4 Block ASSEMBLY

[**ASSEMBLY**]

title title

title Long descriptive title for assembly.

npin num_pins

num_pins The number of rods along the edge of an assembly.

ppitch ppitch

ppitch Cell pitch (cm)

cell cell

cell ??

lattice axial_label, cell_map

axial_label Label for this axial elevation description.

cell_map Lattice map for this axial elevation. Use a dash for an empty location.

axial label, axial_labels, axial_elevations

label Label for this assembly. Label corresponds to assm_map in CORE block.

axial_labels List of axial labels for this assembly description. Corrrespond to labels in lattice maps.

axial_elevations List of axial elevations for this assembly description (cm).

dancoff dancoff_map

dancoff_map Lattice map of dancoff factors (not currently used in the calculation).

grid label, material, mass, height

label Grid label for a single grid type.

material Grid material for this grid type.

mass Grid mass for this grid type (g).

height Grid height for this grid type (cm).

grid_axial grid_map, grid_elev

grid_map List of spacer grid labels for all grids in an assembly (labels correspond to grid card)

grid_elev List of spacer grid elevations for all grids in an assembly (cm). Elevations refer to the grid midpoint.

lower_nozzle lower_nozzle_comp, lower_nozzle_height, lower_nozzle_mass

lower_nozzle_comp Lower nozzle material.

lower_nozzle_height Lower nozzle height (cm)

lower_nozzle_mass Lower nozzle mass (g). Code will calculate the volume of the nozzle given the nozzle mass, and use coolant for remaining volume.

upper_nozzle upper_nozzle_comp, upper_nozzle_height, upper_nozzle_mass

upper_nozzle_comp Upper nozzle material.

upper_nozzle_height Upper nozzle height (cm)

upper_nozzle_mass Upper nozzle mass (g). Code will calculate the volume of the nozzle given the nozzle mass, and use coolant for remaining volume.

fuel fuel

fuel See detailed Materials description.

mat mat

mat See detailed Materials description.

4.5 Block CONTROL

[CONTROL]

title title

title Long descriptive title for control rod description.

npin num_pins

num_pins The number of rods along the edge of an assembly.

stroke stroke, maxstep

stroke Control rod stroke - distance between full-insertion and full-withdrawal (cm)

maxstep Total number of steps between full-insertion and full-withdrawal

cell Cell

Cell See Cell description in ASSEMBLY block

rodmap label, cell_map

label Label for this axial elevation description

cell_map Lattice map for this axial elevation. Use a dash for no control rod.

axial control_label, axial_labels, axial_elevations

control_label Label for this control rod description. Label corresponds to crd_map in CORE block.

axial_labels List of axial labels for this control rod description. Corrrespond to labels in rod maps.

axial_elevations List of axial elevations for this control rod description (cm).

mat mat

mat See detailed Materials description.

4.6 Block INSERT

[INSERT]

title title

title Long descriptive title for assembly insert description.

npin num_pins

num_pins The number of rods along the edge of an assembly.

cell Cell

Cell See Cell description in ASSEMBLY block

rodmap label, cell_map

label Label for this axial elevation description

cell_map Lattice map for this axial elevation. Use a dash for no insert rod.

axial insert_label, axial_labels, axial_elevations

insert_label Label for this assembly insert description. Label corresponds to insert_map in CORE block.

axial_labels List of axial labels for this assembly insert description. Corrrespond to labels in rod maps.

axial_elevations List of axial elevations for this assembly insert description (cm).

mat mat

mat See detailed Materials description.

4.7 Block DETECTOR

[**DETECTOR**]

title title

title Long descriptive title for detector description.

npin num_pins

num_pins The number of rods along the edge of an assembly.

cell Cell

Cell See Cell description in ASSEMBLY block

rodmap label, cell_map

label Label for this axial elevation description

cell_map Lattice map for this axial elevation. Use a dash for no detector rod.

axial detector_label, axial_labels, axial_elevations

detector_label Label for this detector description. Label corresponds to det_map in CORE block.

axial_labels List of axial labels for this detector description. Corrrespond to labels in rod maps.

axial_elevations List of axial elevations for this detector description (cm).

mat mat

mat See detailed Materials description.

4.8 Block EDITS

[EDITS]

axial_edit_bounds axial_edit_bounds

axial_edit_bounds The boundaries of the axial regions over which axial information should be printed.

axial_edit_mesh_delta axial_edit_mesh_delta

axial_edit_mesh_delta Produces a uniform axial output grid (integrates pin powers over a uniform axial mesh).

4.9 Block COBRATF

[COBRATF]

nfuel nfuel

nfuel Number of radial nodes in fuel pellet.

nc nc

nc Conduction option (see CTF manual).

irfc irfc

irfc Friction factor correlation number (see CTF manual).

dhfrac dhfrac

dhfrac Fraction of power deposited directly in coolant.

hgap hgap

hgap Gap conductance (W/m^2 K)

gridloss gridlabel, gridloss

gridlabel Spacer grid label (from ASSEMBLY block).

gridloss Spacer grid loss coefficient.

epso epso

epso Outer iteration convergence criteria.

oitmax oitmax

oitmax Maximum number of outer iterations.

iitmax iitmax

iitmax Maximum number of vessel iterations

dtmin dtmin

dtmin Minimum time step (s).

dtmax dtmax

dtmax Maximum time step (s).

tend tend

tend End of time domain (s).

rtwfp rtwfp

rtwfp Ratio of time step sizes for conduction/fluid.

maxits maxits

maxits Maximum number of iterations.

courant courant

courant Courant limit

maps_filename maps_filename

maps_filename ??

heated_elements_type heated_elements_type

heated_elements_type ??

heater_tube_id heater_tube_id

heater_tube_id ??

heater_tube_od heater_tube_od

heater_tube_od ??

heater_tube_pitch heater_tube_pitch

heater_tube_pitch ??

solver solver

solver ??

parallel parallel

parallel ??

global_energy_balance global_energy_balance

global_energy_balance Convergence criteria

global_mass_balance global_mass_balance

global_mass_balance Convergence criteria

fluid_energy_storage fluid_energy_storage

fluid_energy_storage Convergence criteria

solid_energy_storage solid_energy_storage

solid_energy_storage Convergence criteria

mass_storage mass_storage

mass_storage Convergence criteria

4.10 Block COUPLING

[COUPLING]

epsk epsk

epsk Eigenvalue converence criteria (pcm)

epsp epsp

epsp Power convergence criteria (L2 norm)

eps_temp eps_temp

eps_temp Temperature convergence criteria (degrees F)

eps_boron eps_boron

eps_boron Boron convergence criteria (ppm)

rlx_power rlx_power

rlx_power Power relaxation factor. Recommend 0.5.

rlx_tfuel rlx_tfuel

rlx_tfuel Fuel temperature relaxation factor. Recommend 1.0.

rlx_den rlx_den

rlx_den Density relaxation factor. Recommend 1.0.

maxiter maxiter

maxiter Maximum number of coupled iterations.

read_restart read_restart

read_restart Name of restart file. Leave blank for no restart.

LIME_xml LIME_xml

LIME_xml Name of LIME XML file. Use not recommended.

Chapter 5

Example

This chapter includes a complete input example for a full-core problem. This problem is Problem 7 of the AMA Progression Benchmark Problems and is based upon the publicly available description of the Watts Bar reactors.

```
!
!  Sample Test case for Problem 7 (Full-Core HFP)
!
[CASEID]
  title 'CASL AMA Problem 7 - Watts Bar Unit 1 Cycle 1 - Public'

[STATE]
  power  100.0          ! % of rated power
  flow   100.0          ! % of rated flow
  pressure 2250.0
  feedback on

  tinlet 557.33         ! F - 565K
  tfuel  900.0          ! K     (HFP value)
  boron  1285           ! ppmB
  modden 0.743          ! g/cc
  sym qtr

  rodbank SA 230
          SB 230
          SC 230
          SD 230
           A 230
           B 230
           C 230
           D 167

[CORE]
  size 15               ! assemblies across core
  rated 3411 131.68     ! MW, Mlbs/hr
  apitch 21.5
  height 406.337

  core_shape
    0 0 0 0 1 1 1 1 1 1 1 0 0 0 0
    0 0 1 1 1 1 1 1 1 1 1 1 1 0 0
    0 1 1 1 1 1 1 1 1 1 1 1 1 1 0
    0 1 1 1 1 1 1 1 1 1 1 1 1 1 0
    1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
    1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
    1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
    1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
    1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
    1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
    1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
    0 1 1 1 1 1 1 1 1 1 1 1 1 1 0
```

```
  0 1 1 1 1 1 1 1 1 1 1 1 1 0
  0 0 1 1 1 1 1 1 1 1 1 1 0 0
  0 0 0 0 1 1 1 1 1 1 1 0 0 0 0

assm_map
  1
  2 1
  1 2 1
  2 1 2 1
  1 2 1 2 2
  2 1 2 1 2 3
  1 3 1 3 3 3
  3 3 3 3

insert_map
   -
  20  -
   - 24  -
  20  - 20  -
   - 20  - 20  -
  20  - 16  - 24 12
   - 24  - 16  -  -
  12  -  8  -

crd_map
  1
  - -
  1 - 1
  - - - 1
  1 - - - 1
  - 1 - 1 - -
  1 - 1 - 1 -
  - - - -

crd_bank
  D  -  A  -  D  -  C  -
  -  -  -  -  - SB  -  -
  A  -  C  -  -  -  B  -
  -  -  -  A  - SC  -  -
  D  -  -  -  D  - SA
  - SB  - SD  -  -  -
  C  -  B  - SA  -
  -  -  -  -

det_map
          1 - - 1 - - -
```

```
      1 - - - 1 - - 1 - 1 -
    - 1 - 1 - - 1 - - - - - 1
    - - - - 1 - - 1 - - 1 - -
  1 - - - 1 - - 1 - - 1 - - - 1
  - - - - 1 - 1 - - - - 1 - - -
  - 1 - - - - - - - 1 - 1 - - - 1
  1 - 1 - 1 - 1 - - 1 - 1 1 1 -
  - - - 1 - - 1 - - 1 - - 1 - -
  1 - 1 - - 1 - 1 - - - - - 1 -
  - - - - 1 - - - 1 - 1 - 1 - -
    1 1 - - - - 1 - - - - - -
    - - - - - - 1 - 1 - 1 - 1
      1 - - 1 - 1 - - - - -
          - - 1 - - 1 -

  baffle ss 0.19 2.85
  vessel mod 219.71 cs 241.70

  lower_plate ss  5.0 0.5   ! mat, thickness, vol frac
  upper_plate ss  7.6 0.5   ! mat, thickness, vol frac

  lower_ref  mod 20.0 1.0   ! mat, thickness, vol frac
  upper_ref  mod 20.0 1.0   ! mat, thickness, vol frac

  xlabel  R P N M L K J H G  F  E  D  C  B  A
  ylabel  1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

  mat he     0.0001786
  mat inc    8.19
  mat ss     8.0
  mat zirc   6.56 zirc4

[ASSEMBLY]
  title "Westinghouse 17x17 Assembly"
  npin 17
  ppitch 1.260

  fuel U21 10.257 94.5 / 2.110
  fuel U26 10.257 94.5 / 2.619
  fuel U31 10.257 94.5 / 3.100

  cell 1     0.4096 0.418 0.475 / U21 he zirc
  cell 2     0.4096 0.418 0.475 / U26 he zirc
  cell 3     0.4096 0.418 0.475 / U31 he zirc
  cell 4            0.561 0.602 / mod    zirc       ! guide/instrument tube
  cell 5            0.418 0.475 /     he zirc       ! plenum
```

```
lattice LAT21
     4
     1 1
     1 1 1
     4 1 1 4
     1 1 1 1 1
     1 1 1 1 1 4
     4 1 1 4 1 1 1
     1 1 1 1 1 1 1 1
     1 1 1 1 1 1 1 1 1

lattice LAT26
     4
     2 2
     2 2 2
     4 2 2 4
     2 2 2 2 2
     2 2 2 2 2 4
     4 2 2 4 2 2 2
     2 2 2 2 2 2 2 2
     2 2 2 2 2 2 2 2 2

lattice LAT31
     4
     3 3
     3 3 3
     4 3 3 4
     3 3 3 3 3
     3 3 3 3 3 4
     4 3 3 4 3 3 3
     3 3 3 3 3 3 3 3
     3 3 3 3 3 3 3 3 3

lattice PLEN
     4
     5 5
     5 5 5
     4 5 5 4
     5 5 5 5 5
     5 5 5 5 5 4
     4 5 5 4 5 5 5
     5 5 5 5 5 5 5 5
     5 5 5 5 5 5 5 5 5

axial  1  11.951 LAT21 377.711 PLEN 393.711
```

```
  axial  2  11.951 LAT26 377.711 PLEN 393.711
  axial  3  11.951 LAT31 377.711 PLEN 393.711

  grid END inc  1017 3.866
  grid MID zirc 875  3.810

  grid_axial
      END  13.884
      MID  75.2
      MID 127.4
      MID 179.6
      MID 231.8
      MID 284.0
      MID 336.2
      END 388.2

  lower_nozzle  ss 6.053 6250.0  ! mat, height, mass (g)
  upper_nozzle  ss 8.827 6250.0  ! mat, height, mass (g)

[INSERT]
  title "Pyrex"
  npin 17

  mat pyrx1 2.25 pyrex-vera

  cell 1  0.214 0.231 0.241 0.427 0.437 0.484 / he ss he pyrx1 he ss

  rodmap  PY8
     -
     - -
     - - -
     1 - - -
     - - - - -
     - - - - - 1
     - - - - - - - -
     - - - - - - - - -
     - - - - - - - - - -

  rodmap  PY12
     -
     - -
     - - -
     1 - - -
     - - - - -
     - - - - - -
     - - - 1 - - -
```

```
      - - - - - - - -
      - - - - - - - - -

   rodmap  PY16
      -
      - -
      - - -
      1 - - -
      - - - - -
      - - - - - 1
      - - - 1 - - -
      - - - - - - - -
      - - - - - - - - -

   rodmap  PY20
      -
      - -
      - - -
      1 - - -
      - - - - -
      - - - - - 1
      1 - - 1 - - -
      - - - - - - - -
      - - - - - - - - -

   rodmap  PY24
      -
      - -
      - - -
      1 - - 1
      - - - - -
      - - - - - 1
      1 - - 1 - - -
      - - - - - - - -
      - - - - - - - - -

   axial   8  15.761 PY8  376.441
   axial  12  15.761 PY12 376.441
   axial  16  15.761 PY16 376.441
   axial  20  15.761 PY20 376.441
   axial  24  15.761 PY24 376.441

[CONTROL]
   title "B4C with AIC tips"
   npin 17
   stroke  365.125 230     ! approx for 1.5875 step sizes and 230 max stroke
```

```
  mat aic 10.2
  mat b4c 1.76

  cell 1  0.382 0.386 0.484 / aic he ss
  cell 2  0.373 0.386 0.484 / b4c he ss

  rodmap AIC
     -
     - -
     - - -
     1 - - 1
     - - - - -
     - - - - - 1
     1 - - 1 - - -
     - - - - - - - -
     - - - - - - - - -

  rodmap B4C
     -
     - -
     - - -
     2 - - 2
     - - - - -
     - - - - - 2
     2 - - 2 - - -
     - - - - - - - -
     - - - - - - - - -

  axial  1    17.031
         AIC 118.631
         B4C 377.711

[DETECTOR]
  title "Incore instrument thimble"
  npin 17

  mat he 0.0001786
  mat ss 8.0

  cell 1  0.258 0.382 / he ss

  rodmap  LAT
     1
     - -
     - - -
```

```
    - - - -
    - - - - -
    - - - - - -
    - - - - - - -
    - - - - - - - -
    - - - - - - - - -

  axial 1  0.0 LAT 406.337

[EDITS]
  axial_edit_bounds
       11.951   15.817   24.028   32.239   40.45
       48.662   56.873   65.084   73.295   77.105
       85.17    93.235  101.3    109.365  117.43
      125.495  129.305  137.37   145.435  153.5
      161.565  169.63   177.695  181.505  189.57
      197.635  205.7    213.765  221.83   229.895
      233.705  241.77   249.835  257.9    265.965
      274.03   282.095  285.905  293.97   302.035
      310.1    318.165  326.23   334.295  338.105
      346.0262 353.9474 361.8686 369.7898 377.711

[COBRATF]
  nfuel   3             ! number of fuel rings in conduction model
  nc      1             ! conduction option - radial conduction
  irfc    2             ! friction factor correlation default=2
  dhfrac  0.02          ! fraction of power deposited directly into coolant
  hgap    5678.3        ! gap conductance
  epso    0.001
  oitmax  5
  iitmax  40
  gridloss END 0.9070   ! spacer grid loss coefficient
  gridloss MID 0.9065   ! spacer grid loss coefficient
  dtmin   0.000001
  dtmax   0.1
  tend    0.1
  rtwfp   1000.0
  maxits  10000
  courant 0.8

[COUPLING]
  epsk        5.0  ! pcm
  epsp        1.0e-4
  rlx_power   0.6
  rlx_tfuel   1.0
  rlx_den     1.0
```

```
maxiter     100
```